Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Static friction

Meaning in my words	How can you tell?
How would you explain static friction to a friend?	You push a crate gently and it stays still. What force balances your sideways push?
Example or sketch	**Non-example and why**
A stationary crate on a level floor is pushed right with 3 N below the sliding threshold. Draw friction.	The crate is already sliding across the floor. Would the friction at that contact still be static?

Limit

Meaning in my words	How can you tell?
How would you explain limit to a friend?	The floor can supply up to 6 N. How much friction balances a gentle 2 N horizontal push?
Example or sketch	**Non-example and why**
With a 6 N limit and no other horizontal forces, what do you predict for a 7 N push?	A friend writes 6 N for every trial, even with no sideways push. What does 6 N mean?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Kinetic friction

Meaning in my words How would you explain kinetic friction to a friend?	**How can you tell?** A crate slides right across a stationary floor. Which way does kinetic friction act on the crate?
Example or sketch If the coefficient is 0.2 and the normal force is 10 N, what arrow would you draw?	**Non-example and why** A crate stays put while you push gently. Should you use the kinetic-friction rule?

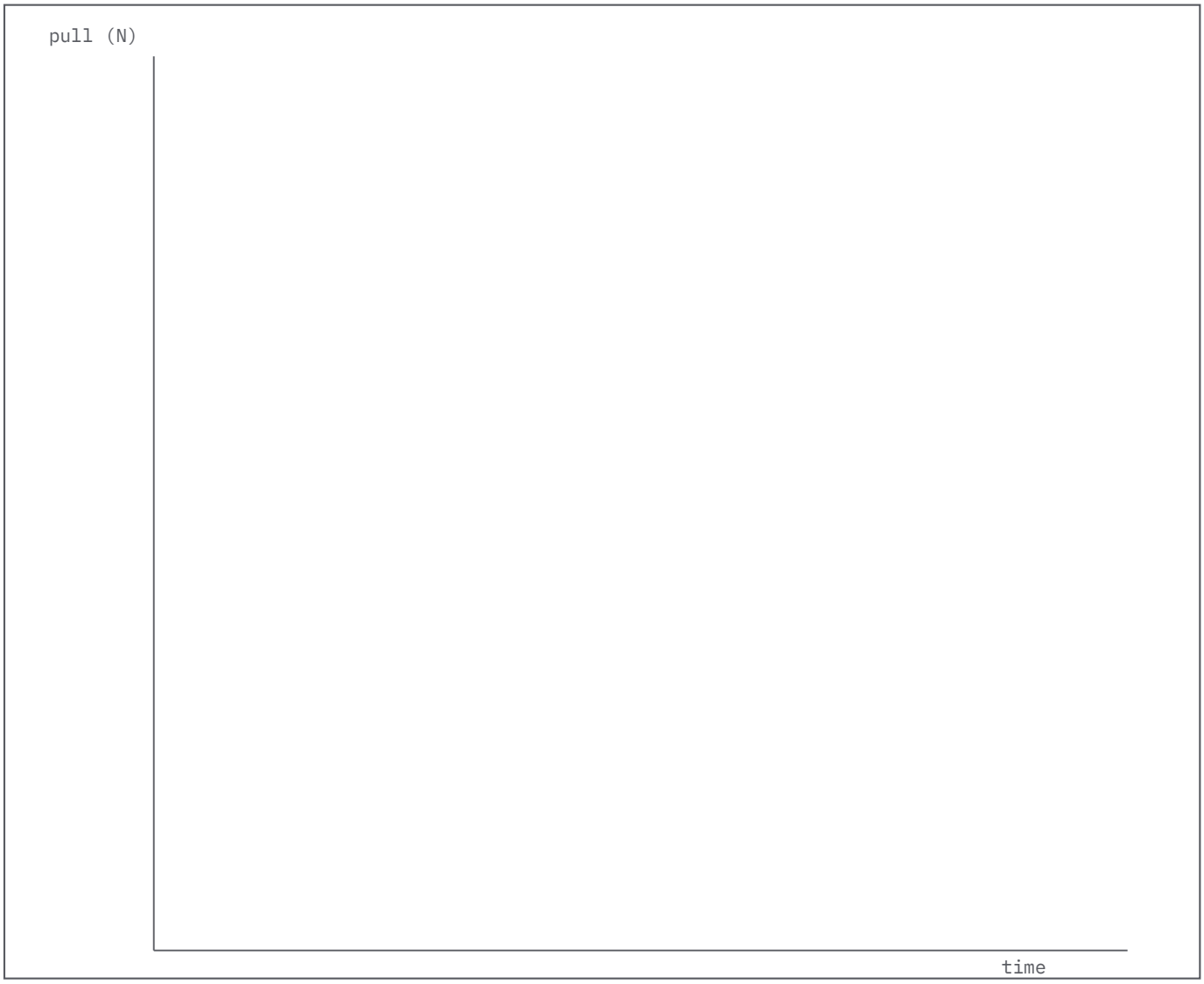
Normal force

Meaning in my words How would you explain normal force to a friend?	**How can you tell?** A crate rests on a ramp. Which way does the ramp push the crate?
Example or sketch How would you draw the normal force on a book on a level table?	**Non-example and why** Someone draws the normal force along the ramp. What would you rotate?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a force trace. Label the peak just before slip and the steady-speed pull while the surfaces slide.



My marks mean

On assignment: Static friction

SECTION E / BACK / FINAL MEETING ONLY Name _____

A show cuts from a block sliding on a rough table to XJ coasting in empty space. The proposed caption says “Both will stop from surface friction.”

PICK A JOB / DELIVER ONE PRODUCT

☐ **Effects engineer**

Specify the friction setting for each scene. Mark the contact surfaces, then tell the animator which motion needs another force to change.

☐ **Science broadcaster**

Record a correction for the cut. Your producer points to the surfaces in each scene and asks what could still bend XJ’s path.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor’s check: Explain how entering an atmosphere would change the force model.